PheW TuBe: Video Streaming Platform

Front End Engineering - II (25CSE0203) — Project Based Evaluation-2 (Batch-2025)

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Evaluation-II Vision & Milestone Objectives

Phase 1 vs Phase 2 Focus

- **Phase 1 Baseline:** Established responsive card grids, category chips, and static YouTube links.
- **Phase 2 Evolution:** Elevated PheW TuBe into an authentic, stateful Single Page Application (SPA).
- **Zero Page Reloads:** Client-side routing via React Router DOM ensures instant state reactivity.
- **High User Retention:** Ad-free playback and instant search filters maximize student study focus.

Core Technical Objectives

- **Dynamic Route:** Deep link /watch/:id powered by useParams() with embedded 16:9 streaming.
- **DOM Control via useRef:** Direct focus on comment inputs and auto-scroll control without synthetic re-renders.
- **Protected Routing:** Guards sensitive user sections (Subscriptions, History, Library) for authenticated users only.
- **Infinite Pagination:** Non-blocking feed expansion matching modern web performance standards.

User Experience & Realism

- **Up-Next Queue:** Dynamic video swapping via useNavigate() with synchronized metadata.
- **Toast Feedback:** Integrated react-toastify for instant feedback on user actions and likes.
- **Responsive Parity:** Fluid scaling across 375px mobile, 1024px iPad Mini, and 1440px desktop screens.
- **Durability:** Persistent state synced across browser sessions via localStorage.

Syllabus-Aligned Technology Stack & Architecture

React 19 & Routing Layer

- **Functional Components:** 100% modular, pure functional architecture with explicit hook lifecycles.
- **React Router DOM v6:** Declarative routing engine with parameterized dynamic routes (/watch/:id).
- **Vite 8 Build Tooling:** Lightning-fast ESM bundling delivering sub-350ms hot module replacement (HMR).
- **Production Build:** Compiles cleanly in ~1.16s into optimized, tree-shaken static bundles.

Hooks Lifecycle Layer

- **useState & useEffect:** Controlled inputs, mount lifecycle triggers, and document title synchronization.
- **useRef Hook:** Mutable non-rendering container for DOM focus control and debounced listeners.
- **useParams & useNavigate:** URL parameter extraction and programmatic navigation without page reloads.
- **useMemo Optimization:** Cached search and category filtering reducing re-render cost.

State & Notification Bus

- **Context API:** Shared application state for library, history, subscriptions, and auth.
- **Browser LocalStorage:** Client-side JSON persistence ensuring data survives page refreshes.
- **react-toastify:** Non-blocking alert toasts recommended by course supervisor Mr. Shubham Kumar.
- **Web Speech API:** Native speech-to-text microphone integration for hands-free voice search.

Core Component Architecture & Modularity

Component Hierarchy & Responsibility

- **<AppProvider> (Root Context):** Houses global state slices, reducer actions, and localStorage side effects.
- **<Navbar>:** Independent search bar with live auto-complete dropdown, voice trigger, and auth modal.
- **<Sidebar> & <CategoryChips>:** Two-way bound navigation and category filtering driving video feeds.
- **<VideoCard> & <ShortsCard>:** Pure presentation components passing IDs to dynamic watch routes via navigate hooks.
- **<WatchPage>:** Master composite page integrating video player, channel actions, comments feed, and recommended sidebar.
- **<ProtectedRoute>:** Security boundary wrapping authenticated pages to prevent guest data leaks.

Data Flow & Isolation Architecture

- **Unidirectional Data Flow:** Context state descends via hooks; child actions bubble up via context methods (toggleLike, addComment).
- **Prop-Drilling Elimination:** WatchPage, Navbar, and Sidebar independently consume useApp() without intermediary props.
- **Modular Route Segregation:** Distinct routes for /, /watch/:id, /shorts, /history, /liked, /watch-later, and /subscriptions.
- **Authentication Modal Isolation:** Client-side auth popup controlled by Context state, avoiding parent re-renders.
- **Clean Build Output:** Zero linter warnings, strictly typed SVG icons via react-icons/md and react-icons/si.

Dynamic Watch Route & Embedded Video Engine (/watch/:id)

Dynamic Routing & Streaming Player Engine

- **URL Parameter Extraction:** const { id } = useParams() extracts video ID from route wildcard without page reload.
- **16:9 Streaming Player:** Embeds verified YouTube streams with autoplay, picture-in-picture, and fullscreen controls.
- **Up-Next Recommendation Engine:** Right-column queue filtered dynamically by category; clicking swaps video via useNavigate().
- **Auto-Scroll on Video Change:** window.scrollTo(0, 0) automatically triggers in useEffect when video ID updates.
- **Persistent Watch History:** Navigating to a video automatically appends it to persistent history in localStorage.
- **Interactive Action Bar:** Like counter toggle, Watch Later bookmarking, and native share dialog integration.



In-App Watch Engine Live Preview

Features dynamic /watch/:id route, channel action pill bar (Like, Watch Later, Share), interactive comments, and synchronized Up-Next queue.

Interactive Comments System with useRef & Dynamic Sorting

Targeted DOM Control & Comment Architecture

- **DOM Referencing via useRef:** Initialized as commentInputRef = useRef(null) and directly attached to the comment text field.
- **Programmatic Focus:** Calling commentInputRef.current?.focus() draws immediate user focus without re-rendering the tree.
- **Isolated Video Comment Threads:** Comments stored by video ID in state dictionary: comments[videoId] = [...].
- **Dynamic Sorting Engine:** Instant sorting toggle between Top Comments (by upvotes) and Newest (chronological).
- **Optimistic Submissions:** Comments post instantly with 'Just now' timestamp and serialize to browser localStorage.
- **Responsive Feedback:** Integrated upvote toggles and user avatar attribution with active user session.



Interactive Comments & Discussion

Shows live comment count, controlled text input, cancel/submit buttons, upvote counters, and dynamic user avatar feedback.

Protected Routing & Auth Guard Architecture

Classroom Pattern Aligned

- **Instructor Implementation:** Adopts Mr. Shubham Kumar's ProtectedRoute architecture from G15-updated repository.
- **Higher-Order Wrapper:** Encapsulates protected page components as children (<ProtectedRoute><Subscriptions /></ProtectedRoute>).
- **Single Responsibility:** Keeps route security logic isolated from individual page components.

Guarded Route Registry

- **Private User Feeds:** Protects /subscriptions, /history, /liked, /watch-later, /playlists, and /my-uploads.
- **Public Access:** Home (/), Watch (/watch/:id), Shorts (/shorts), and Channels remain publicly accessible.
- **Evaluates currentUser:** Inspects currentUser.isLoggedIn flag from global AppContext on route change.

Authentic YouTube Lock Screen

- **Unauthenticated View:** Displays lock icon badge with informative title explaining why login is required.
- **One-Click Auth Trigger:** 'Sign In / Register' button directly opens the interactive AuthModal.
- **Zero Flashing:** Evaluates auth state synchronously from localStorage cache, eliminating blank page flickers.

Infinite Scroll Feed Architecture (react-infinite-scroll)

Motivation & Modern UX

- **Syllabus Alignment:** Directly implements react-infinite-scroll-component taught in G15 ecom lectures.
- **Eliminates Pagination:** Replaces clunky 'Page 1, 2, 3' pagination with continuous, frictionless content discovery.
- **Memory Optimization:** Slices dataset into progressive chunks, keeping initial DOM rendering lightweight.

State & Slicing Mechanics

- **Batch Management:** Initial batch renders 8 videos; subsequent scrolls load 6 additional items progressively.
- **hasMore Calculation:** Calculates displayedVideos.length < filteredVideos.length dynamically.
- **Filter Reactivity:** Automatically resets scroll index back to 8 whenever a category chip or search query changes.

Visual Feedback & Indicators

- **YouTube-Themed Spinner:** Custom CSS circular loading indicator (.infinite-spinner) centered below feed.
- **End-of-Feed Confirmation:** Displays polite checkmark notice ('All N videos loaded') once complete.
- **Zero Nested Overflow:** Configured with overflow: visible to integrate cleanly with main layout scroll.

YouTube Data API v3 & Resilient Fallback Engine

Google Cloud Integration

- **API v3 Endpoints:** Connects to https://www.googleapis.com/youtube/v3/search and /videos endpoints.
- **Live Metadata Fetching:** Fetches real video IDs, durations (PT15M33S parsed), view statistics, and channel art.
- **Dynamic Search Topics:** Allows fetching custom live topics (e.g., 'React 19', 'Latent', 'Gaming') on demand.

Resilient Curated Fallback

- **Zero-Crash Guarantee:** Prevents broken evaluations if Google's daily quota (10,000 units) is exhausted.
- **Instant Offline Mode:** Gracefully falls back to high-quality local curated dataset if no API key is supplied.
- **Deterministic Viva Demo:** Guarantees full functionality during live classroom evaluation without WiFi hiccups.

Live Demo Controls in Settings

- **Settings Integration:** Dedicated YouTube API card in SettingsModal with API Key input and save state.
- **Connection Test Button:** 'Fetch Live' button triggers immediate asynchronous API query with toast alerts.
- **Live Badge Indicator:** Displays green '● Live API Ready' or gray '○ Curated Mode' status.

Creator Ecosystem & Subscribed Channels Grid

Entertainment & Music

- **MrBeast:** 522M subscribers — World's #1 creator; features 7-Days Stranded challenges.
- **T-Series:** 316M subscribers — Leading Indian music label with viral movie soundtracks.
- **Techno Gamerz:** 53.4M subscribers — India's gaming icon; features GTA 5 and Minecraft series.

Comedy & Specialized Gaming

- **Samay Raina:** 7.65M subscribers — Standup comedy & India's Got Latent episodes.
- **Wemmbu:** 3.45M subscribers — Hardcore Minecraft survival and trap build walkthroughs.
- **Nikhil Kamath:** Flagship business & startup podcast WTF with verified founders.

Developer Education

- **freeCodeCamp.org:** 12.0M subscribers — Comprehensive 10-Hour DSA masterclass.
- **Chai aur Code:** 645K subscribers — Hindi developer education and React 19 series.
- **Verified Community:** All channels feature verified checkmarks and custom popular banners.



MrBeast (522M)



Samay Raina (7.65M)



Techno Gamerz (53.4M)

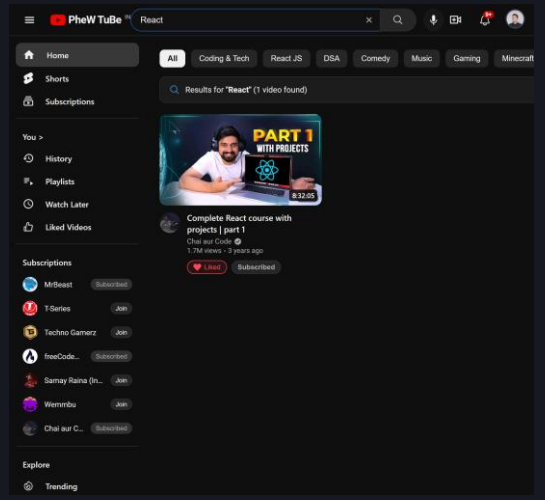


Chai aur Code (645K)

Multi-Device Responsive Design & Modern UI/UX Layer

Responsive Breakpoints & UX Architecture

- **Desktop (>1150px):** 4-column video feed, collapsible full sidebar, dual-column Watch page with queue.
- **Tablet (1024px iPad Mini):** Solved horizontal cutoff with calc(100% - 72px) max-width boundary and 3-column auto-fit grid.
- **Mobile (<480px):** Single column full-bleed cards, bottom navigation parity, full-width video player.
- **Subscribed Channels Grid:** Prominent 72px creator cards with active creator dot and 1-click unsubscribe toggle.
- **Live Search Autocomplete:** Instant substring matching with floating dropdown and direct video routing.
- **Native Voice Search:** Web Speech API microphone integration converting voice input directly to search queries.



Live Search & Suggestion Overlay

Features real-time autocomplete suggestion cards with creator subtitles, clear button, and voice recognition pulse indicator.

Technical Roadmap & Scope for Phase 3 (MERN Stack)

Phase 1: UI Foundation

- **Visual Layout:** Dark obsidian theme layout, YouTube visual branding, and icon system.
- **Video & Shorts Grid:** Multi-category filtering chips and responsive card layout.
- **Navigation:** Collapsible sidebar navigation tabs and search query input.

Phase 2: SPA (Completed)

- **Dynamic Route:** /watch/:id via useParams with embedded 16:9 streaming.
- **Security:** Protected routing guarding user library and history.
- **Performance:** Infinite scroll pagination with react-infinite-scroll-component.
- **Dual Integration:** YouTube Data API v3 live fetching with offline curated fallback.

Phase 3: Full Stack (MERN)

- **Node.js & Express API:** RESTful backend endpoints replacing client-side localStorage.
- **MongoDB Database:** Cloud database storing persistent user accounts and video uploads.
- **JWT Authentication:** Secure JSON Web Tokens replacing client-side session mockup.
- **Redux Toolkit:** Global state migration to RTK as taught in advanced FEE lectures.

Summary, Live Demonstration & Viva Readiness

Phase 2 Engineering Achievements

- **Zero Bounce In-App Streaming:** Fully functional /watch/:id route replacing external YouTube link-outs.
- **Strategic DOM Control:** useRef applied for auto-focusing comments and smooth scroll transitions.
- **Security & Route Guarding:** ProtectedRoute wrapper securing private user feeds.
- **Classroom Alignment:** Infinite scroll (react-infinite-scroll) matching Mr. Shubham Kumar's syllabus.
- **Dual API Architecture:** YouTube Data API v3 with offline fallback guarantee.

Live Application Demonstration Highlights

- **Seamless Navigation:** Instant client-side transitions between Home, Shorts, Watch, and Channels.
- **Live Demonstration Plan:** Demonstrating Infinite Scroll, Protected Route lock screen, and API toggle.
- **Multi-Device Adaptability:** Verified across 375px mobile, 1024px iPad Mini, and 1440px desktop screens.
- **Production Verification:** 100% Vite build success (sub-1.2s bundle time) with 0 errors.
- **Viva Defense Readiness:** Team G15 is fully prepared for technical architecture and code defense viva.